In this case, either some iterate of $\tau \circ \sigma$ has significant contraction, or the short geodesics cycle and M/τ is toroidal.

§2 assembles the results on Teichmüller theory and geometric limits of quadratic differentials developed in [Mc1]. §3 discusses hyperbolic 3-manifolds and the skinning map. In §4 we prove a compact convex acylindrical hyperbolic 3-manifold can be deformed so its boundary becomes totally geodesic; the proof is short and it gives a good introduction to the general gluing problem. §5 derives the dependence on moduli, and §6 establishes uniform contraction for acylindrical manifolds. It is here we analyze short geodesics. §7 treats cylindrical manifolds and completes the proof of Theorem 1.2.

Remarks

1. The proof of Theorem 1.2 parallels similar approaches to the mapping class group and to rational maps.

Any iteration f on Teichmüller space, such that the distance from X to $f(X)$ remains bounded as $[X]$ tends to infinity in moduli space, preserves a system of short geodesics on X.

Thus a mapping class whose minimal translation distance is not achieved is reducible.

For critically finite branched covers of the sphere, the amount of contraction of f is controlled by the location of X in moduli space. Hence either $[f^k(X)]$ remains in a compact set—and one constructs a rational map—or one finds a system of annuli providing an obstruction to rationalization.

In the case of 3-manifolds, either one obtain a hyperbolic structure, or one finds a system of simple closed curves bounding cylinders which link up to form a nonperipheral torus. This torus provides a topological obstruction to geometrization.

2. To give some idea of other approaches to the skinning map, we focus on the acylindrical case.

Thurston has shown that the space $AH(M)$ of *all* hyperbolic structures on an acylindrical manifold is compact in the algebraic topology [Th1]. This result has also been obtained using trees and valuations by Morgan and Shalen [MS1, MS2]. To obtain contraction, one first extends the skinning map to $AH(M)$, then uses the theory of geometrically tame ends to show the image lies entirely in $QH_0(M)$, the space of hyperbolic structures quasi-isometric to the initial one N. It follows that σ has a bounded image in Teichmüller space, hence is uniformly contracting.

3. Our techniques do not yield the bounded image theorem, nor do they apply to the case $M = S \times [0, 1]$ where S is a surface. This case arises naturally for 3-manifolds which fiber over the circle.

Acknowledgements. The connection between Kra's Θ conjecture and the skinning map was first pointed out to me many years ago by John Hubbard. I'd also like to thank Bill Thurston for explaining much of his original proof, Andrew Casson, Howard Masur and Peter Shalen for suggesting revisions, and Ecole Normale Supérieure, IHES and MSRI for their hospitality while this work was being completed.

2. Riemann surfaces and quadratic differentials

2.1. The infinitesimal form of the Teichmüller metric

Let X be a hyperbolic Riemann surface with finitely generated fundamental group. Each end of X is either a finite volume cusp, or an infinite volume funnel.

The quasiconformal type of X can be encoded by a *pared surface* (S, P), i.e. a compact oriented surface S with specified *parabolic locus* $P \subset \partial S$ designating those components of the boundary corresponding to parabolic cusps. The interior of a pared surface carries a unique quasiconformal structure with infinite modulus ends at P and finite modulus ends at $\partial S - P$.

The *Teichmüller space* Teich (S, P) parameterizes Riemann surfaces X marked by quasiconformal maps $\phi : \text{int}(S) \to X$ up to isotopy pointwise fixing the ideal boundary. We allow the shorthand Teich(S) when $P = \partial S$.

The Teichmüller metric is given by

$$d(X_1, X_2) = \inf \log K(\phi_1 \circ \phi_2^{-1})$$

where the infimum is over representative markings ϕ_i and K denotes maximal dilatation.

This metric can be described on an infinitesimal level as follows.

Let $Q(X)$ denote the Banach space of holomorphic quadratic differentials $\phi(z)dz^2$ on X satisfying

$$\|\phi\| = \int_X |\phi(z)||dz|^2 < \infty \ .$$

$Q(X)$ is the cotangent space to Teichmüller space at X, and the norm above is the Teichmüller cometric.

There is a natural pairing between $Q(X)$ and the Banach space $M(X)$ of measurable Beltrami differentials $\mu(z)d\bar{z}/dz$ satisfying

$$\|\mu\| = \text{ess sup}\,|\mu(z)| < \infty \ ,$$

namely

$$\langle \phi, \mu \rangle = \text{Re} \int_X \phi(z)\mu(z)|dz|^2 \ .$$

Geometrically, $\langle \phi, \mu \rangle$ measures the extent to which the linefields of ϕ and μ are synchronized, averaged with respect to the area form $|\phi\mu|$.

The quotient $M(X)/Q(X)^{\perp}$ is the tangent space to Teichmüller space at X, and the quotient norm is the infinitesimal form of the Teichmüller metric.

This means a Beltrami differential μ gives an infinitesimal quasiconformal deformation of X whose length in the Teichmüller metric is

$$l(\mu) = \sup\{\langle \phi, \mu \rangle : \|\phi\| = 1\} \ .$$

There are many μ of length zero, giving infinitesimal maps of X to itself. Frequently $l(\mu) < \|\mu\|$, indicating that μ can be replaced by an equivalent map of lesser distortion.

See [Gar] for more details on the analytic foundations of Teichmüller theory.